Dumbbell with viscosity — Solution

X26 (2026) — English translation

A1 — 0.60 pts

Find the distance x_1 from the end of the dumbbell m_1 to the attachment point.

With a fixed orientation of the rotation axis relative to the dumbbell, the Huygens-Steiner theorem can be used to minimize the moment of inertia: for the rotation axis passing through an arbitrary point P, the moment of inertia J_{P} is related to the moment of inertia J_{O}, where O is the center of mass of the system, as

$$J_{\mathrm{P}} = J_{\mathrm{O}} + M \cdot \mathrm{OP}^2.$$

Thus, in the case of минимального J_{P} true $\mathrm{P} = \mathrm{O}$. For вычисления расстояния от края m_2 we use the definition of the center of mass:

Answer: \textbfAnswer:

$$x_2 = \frac{m_1}{m_1 + m_2} l.$$

A2 — 0.20 pts

Find the moment of inertia J.

Let's find the distance x_1 from the heavy end of the dumbbell to its center of mass:

$$x_1 = \frac{m_2}{m_1 + m_2} l.$$

Let's use the definition of the moment of inertia:

Next, for convenience, we introduce the reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$. Then

$$J_{\mathrm{O}} = \mu l^2.$$

Answer: \textbfAnswer:

$$J_{\mathrm{O}} = m_1 x_1^2 + m_2 x_2^2 = \frac{m_1 m_2}{m_1 + m_2} l^2.$$

A3 — 0.30 pts

Calculate the angular acceleration ε of the dumbbell at time $t = 0$.

In the case of a fixed axis of rotation, we can write the equation of rotational motion in the form

$$J\vec{\varepsilon} = \sum_i \vec{M}_i.$$

In the case of плоского движения направления momenta сил and углового ускорения очевидны, так что next перепишем equation в скалярном виде. Согласно условию, внешний вынуждающий moment постоянен and equals $M_{\text{outer}} = F \cdot x_1$, a for/at нулевой угловой скорости friction force отсутствует. Substituting expression for x_1, we have

Answer: \textbfAnswer:

$$\varepsilon = \frac{m_2}{m_1 + m_2} \frac{Fl}{\mu l^2} = \frac{F}{m_1 l}.$$

A4 — 0.60 pts

Calculate the steady-state angular velocity ω_{lim} of the dumbbell rotation after a long time.

The frictional moment M_{fric} arises due to the linear velocities $v_1,\ v_2$ of the two point ends of the rod:

$$v_1 = \omega x_1;\quad v_2 = \omega x_2 \ \Rightarrow\ f_1 = -\gamma\omega x_1;\quad f_2 = -\gamma\omega x_2.$$

Очeit can be seen that moment направлен против вынуждающего постоянного momenta. Let us write его:

$$M_{\text{fric}} = f_1 x_1 + f_2 x_2 = -\gamma(x_1^2 + x_2^2)\omega.$$

Now we can write down the equation of rotational motion at an arbitrary moment of time:

$$\mu l^2 \varepsilon = \mu l^2 \frac{\mathrm{d}\omega}{\mathrm{d}t} = F x_1 - \gamma(x_1^2 + x_2^2)\omega \ \Leftrightarrow\ \mu l^2 \frac{\mathrm{d}\omega}{\mathrm{d}t} + \gamma(x_1^2 + x_2^2)\omega = F x_1.$$

Solution этого уравнения givesся следующей формулой:

$$\omega(t) = \frac{F x_1}{\gamma(x_1^2 + x_2^2)}\left(1 - e^{-\frac{\gamma(x_1^2+x_2^2)}{\mu l^2}t}\right) = \frac{F}{\gamma}\frac{m_2(m_1+m_2)}{(m_1^2+m_2^2)l}\left(1 - e^{-\frac{\gamma(m_1^2+m_2^2)}{(m_1+m_2)m_1 m_2}t}\right).$$

Next let us find ω_{lim} как limit $\omega(t)$ for/at $t \to \infty$, or from дифференциального уравнения с conditionм $\dot{\omega} = 0$.

Answer: \textbfAnswer:

$$\omega_{\text{lim}} = \lim_{t\to\infty} \omega(t) = \frac{m_2(m_1+m_2)}{m_1^2+m_2^2}\frac{F}{\gamma l}.$$

A5 — 0.50 pts

Find the moment of time τ at which the angular velocity is $\omega_{lim}/2$.

According to the explicit formula $\omega(t)$, obtained by solving the previous paragraph, we have the answer for the characteristic time τ (the time during which the exponent will significantly change its value).

Answer: \textbfAnswer:

$$\tau = \frac{(m_1+m_2)\cdot m_1 m_2}{\gamma(m_1^2+m_2^2)}\ln(2).$$

B1 — 0.20 pts

Calculate the angular acceleration ε of the dumbbell at time $t = 0$.

The equation of rotational motion of a system with mass m for an axis moving translationally looks like

$$J\vec{\varepsilon} = \vec{M}_{\text{outer}} + m\left[\vec{v}_c, \vec{v}_{\text{P}}\right],$$

where $\vec{v}_{\text{P}}$ — (поступательная) velocity оси, а $\vec{v}_c$ — velocity of center of mass. Thus, for движущейся оси, совпадающей с центром масс, equation движения так же for/atнимает вид

$$J\vec{\varepsilon} = \vec{M}_{\text{outer}},$$

and the answer for the angular acceleration of the previous part of the problem remains valid.

Answer: \textbfAnswer:

$$\varepsilon = \frac{F}{m_1 l}.$$